

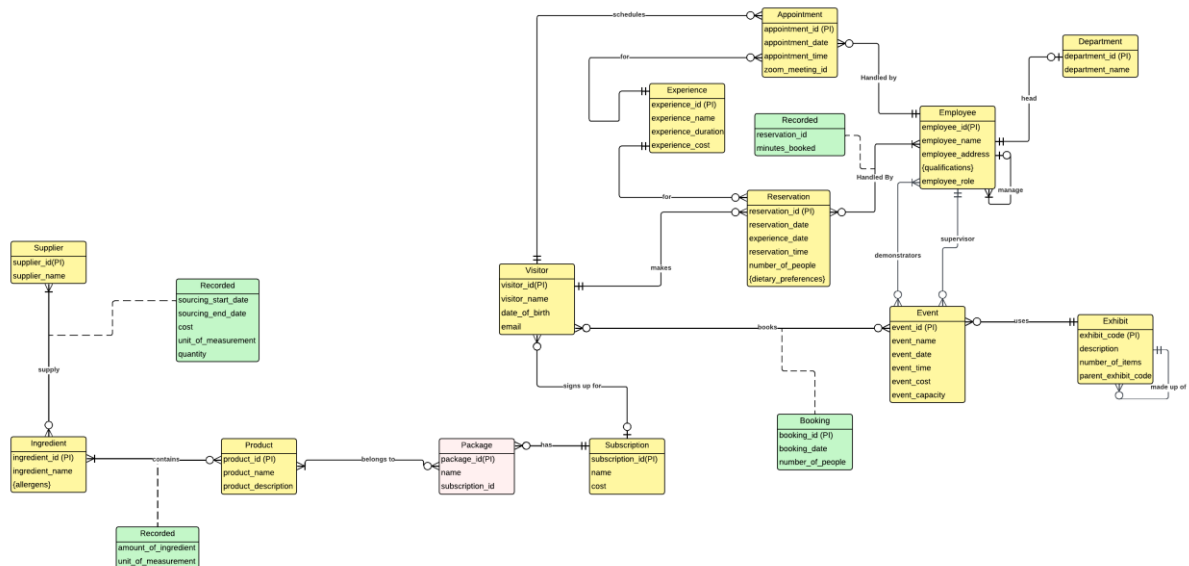
COMP1350 Assignment

Group Number:

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Solutions:

Task 1:



Task 2: ANSFORMATION

Step 1	Visitor: visitor_id (PK), visitor_name, date_of_birth, email Event: event_id(PK), event_name, event_date, event_time, event_cost, event_capacity Experience: experience_id (PK), experience_name, experience_duration, experience_cost Exhibit: exhibit_code (PK), description, numbers_of_items, parent_exhibit_code Subscription: subscription_id (PK), name, cost Ingredient: ingredient_id (PK), ingredient_name
Strong Entity	

	Product: product_id(PK), product_name, product_description Employee: employee_id (PK), employee_name, employee_address, employee_role Department: department_id (PK), department_name Supplier: supplier_id (PK), supplier_name
Step 2 Weak Entity	Package: subscription_id (PK,FK), package_id (PK), name
Step 3 One to One relationship	Department: department_id(PK), department_name, manager_id(FK)
Step 4 One to many relationship	Ingredient: ingredient_id (PK), ingredient_name, supplier_id(FK) Visitor: visitor_id (PK), visitor_name, date_of_birth, email, subscription_id(FK) EmployeeManageOthers: employee_id (PK), employee_name, employee_address, employee_role, manager_id(FK) ExhibitsMadeUpOfOthers: exhibit_code (PK), description, numbers_of_items, parent_exhibit_code (FK)
Step 5 Many to many relationship	ProductBelongstoPackage: package_id(PK,FK), product_id(PK,FK) IngredientsContainInProduct: ingredient_id (PK,FK), product_id(PK,FK), amount_of_ingredient, unit_of_measurement VisitorBookingEvent: visitor_id (PK,FK), event_id (PK,FK), booking_id (PK), booking_date, number_of_people IngredientSourcing:supplier_id (PK,FK), ingredient_id (PK,FK), sourcing_start_date, sourcing_end_date, cost, unit_of_measurement, quantity

	ReservationHandledByEmployee: employee_id (PK, FK), reservation_id (PK, FK), minutes_booked
Step 6 Multi-valued attribute	Employee_Qualifications: employee_id (PK, FK), qualification (PK) Reservation_Dietary_Preferences: reservation_id (PK, FK), dietary_preference (PK) Ingredient_Allergens : ingredient_id (PK, FK), allergens (PK)
Step 7 Ternary Relationship	Reservation: experience_id(FK), visitor_id (FK), employee_id(FK), reservation_id (PK), reservation_date, experience_date, reservation_time, number_of_people Appointment: visitor_id (FK), employee_id (FK), experience_id (FK), appointment_id (PK), appointment_date, appointment_time, zoom_meeting_id Event: visitor_id (FK), exhibit_id (FK), employee_id (FK), event_id(PK), event_name, event_date, event_time, event_cost, event_capacity

Final Lists Of Tables:

Visitor: visitor_id (**PK**), visitor_name, date_of_birth, email, subscription_id(**FK**)

Event: visitor_id (**FK**), exhibit_id (**FK**), employee_id (**FK**), event_id(**PK**), event_name, event_date, event_time, event_cost, event_capacity

Reservation: experience_id(**FK**), visitor_id (**FK**), employee_id(**FK**), reservation_id (**PK**), reservation_date, experience_date, reservation_time, number_of_people

Appointment: visitor_id (**FK**), employee_id (**FK**), experience_id (**FK**), appointment_id (**PK**), appointment_date, appointment_time, zoom_meeting_id

Ingredient: ingredient_id (**PK**), ingredient_name, supplier_id(**FK**)

Department: department_id(**PK**), department_name, manager_id(**FK**)

Package: subscription_id (**PK,FK**), package_id (**PK**), name

Experience: experience_id (**PK**), experience_name, experience_duration, experience_cost

Exhibit: exhibit_code (**PK**), description, numbers_of_items, parent_exhibit_code

Subscription: subscription_id (**PK**), name, cost

Employee: employee_id (**PK**), employee_name, employee_address, employee_role

Supplier: supplier_id (**PK**), supplier_name

EmployeeManageOthers: employee_id (**PK**), employee_name, employee_address, employee_role, manager_id(**FK**)

ExhibitsMadeUpOfOthers: exhibit_code (**PK**), description, numbers_of_items, parent_exhibit_code (**FK**)

Employee_Qualifications: employee_id (**PK, FK**), qualification (**PK**)

Reservation_Dietary_Preferences: reservation_id (**PK, FK**), dietary_preference (**PK**)

Ingredient_Allergens : ingredient_id (**PK, FK**), allergens (**PK**)

ProductBelongstoPackage: package_id(**PK,FK**), product_id(**PK,FK**)

IngredientsContainInProduct: ingredient_id (**PK,FK**), product_id(**PK,FK**), amount_of_ingredient, unit_of_measurement

VisitorBookingEvent: visitor_id (**PK,FK**), event_id (**PK,FK**), booking_id (**PK**), booking_date, number_of_people

Ingredient Sourcing: supplier_id (**PK,FK**), ingredient_id (**PK,FK**), sourcing_start_date, sourcing_end_date, cost, unit_of_measurement, quantity

ReservationHandledByEmployee: employee_id (**PK, FK**), reservation_id (**PK, FK**), minutes_booked